MAHATMA GANDHI UNIVERSITY, KOTTAYAM

B.Sc. (Hons) CHEMISTRY – Semester IV

Model Question Paper

Organic Chemistry – II (MG4DSCCHE200)

Time: 3 Hours Maximum Marks: 50

PART A – Multiple Choice Questions

Answer all questions. (10 × 1 = 10 marks)

1. Which of the following alcohols is produced when Ethyl Magnesium Bromide reacts with Formaldehyde?

   a) Ethanol b) Propan-1-ol c) Propan-2-ol d) Methanol **(CO1, Apply)**

2. What is the primary product formed when a primary alcohol is reacted with alkaline $KMnO_4$

   a) Ketone b) Aldehyde c) Carboxylic acid d) Alkane **(CO1, Remember)**

3. Intermolecular hydrogen bonding in *para*-nitrophenol, compared to *ortho*-nitrophenol, leads to:

   a) Higher boiling point b) Lower solubility in water c) Lower boiling point d) Increased volatility **(CO1, Understand)**

4. Which of the following compounds is also known as 2,4,6-trinitrophenol?

   a) Catechol b) Resorcinol c) Quinol d) Picric acid **(CO1, Remember)**

5. Which carbonyl compound is commonly used as a preservative for biological specimens (formalin)?

   a) Acetone b) Benzaldehyde c) Formaldehyde d) Acetaldehyde **(CO2, Remember)**

6. The reaction of an aldehyde with a primary amine $R\text{-}NH_2$ results in the formation of: a) Oxime b) Schiff's base c) Hydrazone d) Phenylhydrazone **(CO2, Understand)**

7. The Perkin condensation is used specifically for the synthesis of:

   a) alpha, beta-unsaturated aromatic acids b) Saturated ketones c) beta-hydroxy aldehydes d) Cyclic ethers **(CO2, Apply)**

8. Which carboxylic acid lacks an alpha-hydrogen and exhibits some properties similar to aldehydes, such as reducing Tollen's reagent?

   a) Acetic acid b) Formic acid c) Benzoic acid d) Oxalic acid **(CO3, Understand)**

9. The Soda-lime decarboxylation of Sodium Acetate produces:

   a) Ethane b) Methane c) Acetic acid d) Methanol **(CO3, Apply)**

10. The Reformatsky reaction involves the reaction of an aldehyde or ketone with:

    a) alpha-halo ester and Zinc metal b) Grignard reagent c) Ammonia and Hydrogen d) Acid chloride and Sodium **(CO3, Remember)**

PART B – Short Answer Questions

Answer any FOUR. (4 × 5 = 20 marks)

11. Discuss the metabolism of Ethanol in the human body and the role of the enzyme alcohol dehydrogenase. **(CO1, Understand)**

12. Write the mechanism for the Nitration of Phenol. Why does phenol undergo nitration more easily than benzene? **(CO1, Analyse)**

13. Explain the Knoevenagel reaction including the reagents used and the type of product formed. **(CO2, Apply)**

14. Outline the mechanism for the Beckmann rearrangement of an oxime to an amide. **(CO2, Analyse)**

15. Compare the acid strength of Benzoic acid, *p*-nitrobenzoic acid, and *p*-methoxy benzoic acid. Justify your answer. **(CO3, Analyse)**

16. Give the structures and industrial uses each for Oxalic acid, Phthalic acid, and salicylic acid. **(CO3, Remember)**

PART C – Short Essay Questions

Answer any TWO. (2 × 10 = 20 marks)

17. Describe the chemical methods used for the "Ascent" and "Descent" of the alcohol homologous series with suitable examples. **(CO1, Remember)**

18. Provide detailed mechanisms and salient features of the Benzoin condensation and Claisen-Schmidt reaction **(CO2, Remembet)**

19. Detail the chemical pathways required to convert Acetic acid to Propionic acid (Ascent) and Propionic acid to Acetic acid (Descent). **(CO3, Apply)**

20. A chemist proposes using an amide as the starting material to synthesize an acid chloride in a single-step nucleophilic substitution. Evaluate the feasibility of this synthetic route based on the relative nucleophilicity of acyl derivatives and the stability of the leaving groups. If the route is inefficient, suggest and justify an alternative for interconverting these two functional groups. **(CO3,Evaluate)**